MODULE 01 OF 03 · 30 MINUTES

Software Development Life Cycle …

How an idea becomes working software — and where testing fits

OBJECTIVE 1

Explain what the SDLC is and why teams use it

OBJECTIVE 2

Name the six phases and who works in each

OBJECTIVE 3

Show where QA adds value in every phase

4 min

What is the SDLC, and why does it matter?

The **Software Development Life Cycle** is the step-by-step process teams follow to **plan, design, build, test, release and maintain** software.

A shared roadmap

Everyone knows what happens next and who owns it.

Quality built in

Checks happen at every step, not only at the end.

Time and cost control

Clear phases make plans and progress measurable.

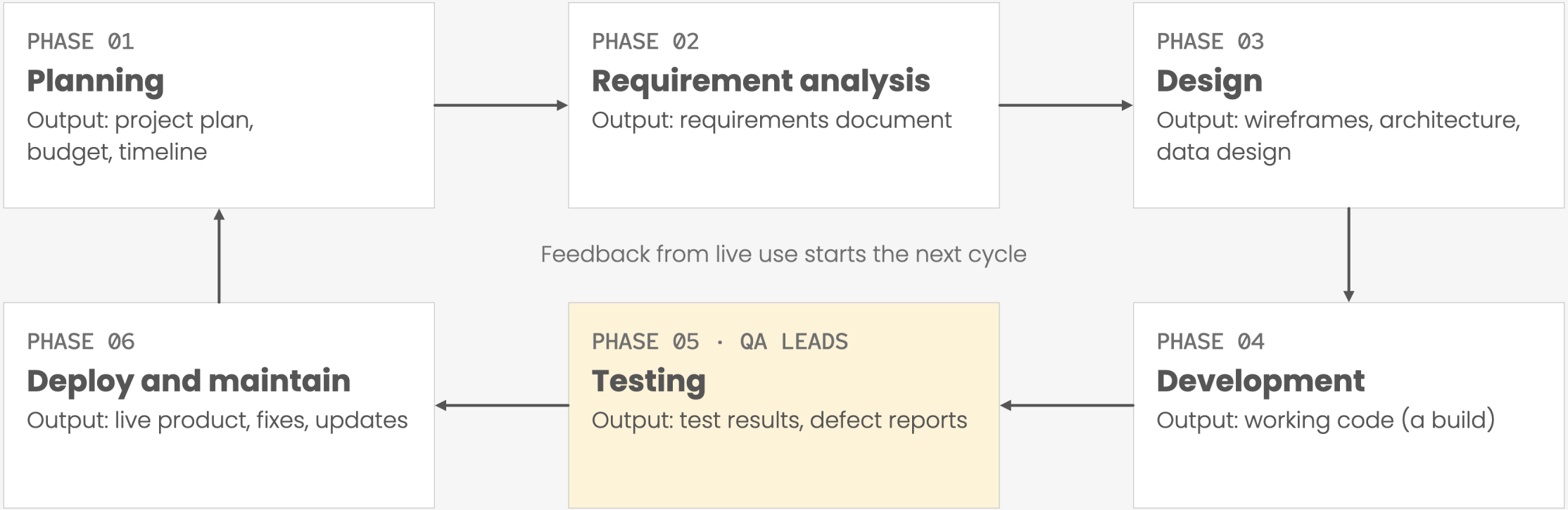
Fewer surprises

Defects found early are cheaper to fix than late ones.

Analogy: building a house. Budget, owner's wish list, architect's drawings, construction, safety inspection, move-in and repairs.

3 min

The six phases form a cycle



3 min

Plan it, define it, design it

01 Planning

What: decide goals, scope, budget, timeline and team.

Output: project plan

> **ShopEasy:** launch an electronics store app in six months with one product team.

QA: estimate test effort; plan tools and test environments.

02 Requirement analysis

What: gather and write down what the software must do.

Output: requirements document

> **ShopEasy:** shoppers must search, add to cart and pay by card or UPI.

QA: review requirements; ask questions; spot gaps.

03 Design

What: decide how it will work — screens, architecture, data.

Output: wireframes, design documents

> **ShopEasy:** wireframes for the cart page; tables for products and orders.

QA: draft test scenarios from the designs.

3 min

Build it, test it, release and care for it

04 Development

What: write the code; unit-test and review it.

Output: a working build

ShopEasy: developers build search, cart and checkout features.

QA: write test cases; prepare test data; try early builds.

05 Testing

What: check the software against the requirements.

Output: test results, defect reports

ShopEasy: confirm a failed payment never creates an order.

QA: run tests; log defects; retest fixes; run regression.

06 Deploy and maintain

What: release to users; then fix, update and improve.

Output: live product, patches

ShopEasy: the app goes live; a coupon bug is fixed in an update.

QA: smoke test the release; regression test every fix.

3 min

Who does what in the SDLC

Role	Main job	Most active in	ShopEasy example
Business analyst	Understands business needs; writes clear requirements	Requirements	Writes “Shoppers can pay with UPI”
Designer (UI/UX)	Designs screens and user journeys that are easy to use	Design	Lays out the checkout page
Developer	Writes, reviews and unit-tests the code	Development	Builds the Add to Cart feature
QA tester	Checks the product meets requirements; finds defects early	Every phase, mainly Testing	Tests checkout with valid and invalid cards

You will also meet project managers, product owners, architects and DevOps engineers. Titles vary by company; the jobs stay the same.

3 min

Where does QA fit? In every phase.

01	02	03	04	05	06
Planning	**Requirements**	**Design**	**Development**	**Testing**	**Deploy and maintain**
Estimate test effort; plan tools and environments	Review requirements; ask questions; spot gaps	Draft test scenarios; plan test data	Write test cases; try early builds	Run tests; log and retest defects; regression	Smoke test releases; test every fix

Shift left: involve QA early. Phase 5 is where QA runs tests; every other phase is where QA prevents defects.

QA's own steps — from test planning to test closure — are called the Software Testing Life Cycle (STLC). You will study it later.

5 min

Phase sort

In pairs, match each situation to its SDLC phase. Three minutes to discuss, two to review.

1. The team agrees the launch date and budget.
2. A BA interviews sales staff about payment options.
3. A designer sketches the checkout screen.
4. A developer codes the wishlist button.
5. QA finds the discount is applied twice.
6. The app is published on the app stores.
7. A customer reports a bug; a fix is released.
8. QA asks, “What if the card payment fails?”

ANSWER KEY

1 Planning

2 Requirement analysis

3 Design

4 Development

5 Testing

6 Deployment

7 Maintenance

8 Requirement analysis

No. 8 is the trap: QA questions belong early, before any code exists.

3 min

Quick quiz: show A, B, C or D

QUESTION 1

Which phase gathers and documents what the software must do?

A Planning

B Requirement analysis

C Design

D Testing

QUESTION 2

When should QA first get involved in a project?

A After coding is finished

B Only during deployment

C From requirements onward

D Only when bugs appear

QUESTION 3

Drawing checkout wireframes belongs to which phase?

A Planning

B Development

C Design

D Maintenance

B — Requirement analysis

This is where the team defines what to build.

C — From requirements onward

Early reviews catch problems while fixes are cheap.

C — Design

Wireframes decide how the screens will look and work.

1 min

Five things to remember

01 The SDLC is the step-by-step path from idea to working, maintained software.

02 Six phases: planning, requirements, design, development, testing, deployment and maintenance.

03 Each phase produces an output that feeds the next one.

04 QA adds value in every phase; defects found early are cheaper to fix.

05 Testing checks the product against the requirements.

Module 01 complete

- ✓ What the SDLC is and why
- ✓ The six phases and outputs
- ✓ Roles in the team
- ✓ Where QA fits

Up next: Agile and Scrum

MODULE 02 OF 03 · 30 MINUTES

Agile methodology and Scrum …

How modern teams organise and deliver work in short, repeatable loops

OBJECTIVE 1

Explain Agile, and how it differs from Waterfall

OBJECTIVE 2

Explain Scrum: roles, sprints, events, backlogs

OBJECTIVE 3

Describe how QA works inside an Agile team

3 min

What is Agile, and why do teams use it?

Agile is a way of thinking about software work: build in small pieces, get feedback early, and adapt to change. It is a set of values and principles — not one fixed process.

WHY TEAMS USE IT

Requirements change, so plans must be able to change too.

Users see working software early and give feedback.

Problems surface in weeks, not after a year.

AGILE PRINCIPLES IN PLAIN WORDS

People and teamwork matter more than rigid tools.

Working software matters more than long documents.

Work with the customer, don't only negotiate contracts.

Respond to change instead of following a fixed plan.

Deliver often, in short cycles. Reflect and improve.

3 min

Waterfall delivers once. Agile delivers in loops.

Waterfall

Plan	Design	Build	Test

One long pass. Users see the product only at the end.

Agile (iterative and incremental)

Loop 1	Loop 2	Loop 3	Loop 4

Each loop plans, builds, tests and releases a small working piece.

Aspect	Waterfall	Agile
Change	Hard and costly once started	Welcomed at every loop
Delivery	One big release at the end	Small working releases often
Testing	Mostly after development	Continuous, inside every loop
Customer	Involved at start and end	Involved throughout

Iterative = repeat and improve. **Incremental** = add working pieces. Waterfall still suits fixed, well-understood work.

3 min

Agile is the mindset. Scrum is one way to practise it.

AGILE

Values and principles

Says what matters. Gives no fixed steps or roles.

SCRUM

A framework that follows Agile

Defines roles, sprints, events and backlogs.

OTHERS

Kanban, XP and more

Also Agile, but with different practices.

THE THREE SCRUM ROLES

☆

Product Owner

Owns the product vision and the Product Backlog. Decides what is built first. ShopEasy: picks Search before Wishlist.

Scrum Master

Coaches the team in Scrum and removes blockers. A servant leader, not a boss. ShopEasy: gets a test server set up.

</>

Developers

Everyone who builds the product, including QA testers. They deliver a working, tested increment each sprint.

4 min

One sprint, four events

A **Sprint** is a fixed time box of one month or less (often two weeks) that ends with a working, tested increment.

DAY 1

Sprint Planning

Team picks backlog items and agrees a Sprint Goal.

EVERY DAY

Daily Scrum

15 minutes. Plan the next 24 hours and flag blockers.

LAST DAY

Sprint Review

Show the working increment; collect stakeholder feedback.

AFTER REVIEW

Retrospective

Team reflects on how it worked and picks improvements.

Then the next sprint starts straight away. The review looks at the product; the retrospective looks at the team's way of working.

4 min

ShopEasy delivered over several sprints

Product Backlog
Ordered list of everything the product may need. Owned by the Product Owner. Always changing.

Sprint Backlog
Items chosen for this sprint plus the team's plan to deliver them. Owned by the Developers.

Sprint	Sprint goal	Working increment	QA focus
Sprint 1	Shoppers can find products	Search and product page	Search tests
Sprint 2	Shoppers can sign in	Registration and login	Login tests
Sprint 3	Shoppers can collect items	Add to cart	Cart and regression
Sprint 4	Shoppers can buy	Checkout and payment	Payment tests
Sprint 5	Shoppers can track orders	Order tracking	Full regression

3 min

QA works inside the sprint, not after it

BEFORE THE SPRINT

Refine

Review stories and acceptance criteria. Ask “what if?” questions. Estimate testing effort.

DURING THE SPRINT

Test as you build

Write test cases while developers code. Test each story the moment it is ready. Log defects and retest fixes.

END OF SPRINT

Confirm done

Run regression tests. Take part in the review. Raise testing issues in the retrospective.

Definition of Done: a shared checklist. A story is only done when coded, reviewed, tested and accepted.

Whole-team quality: developers write unit tests; QA brings the tester’s view. Quality belongs to everyone.

4 min

Plan a sprint

In groups of three, play Product Owner, Scrum Master and Developer. The ShopEasy Product Bac ek sprint.

Product search	Small
Login	Small
Add to cart	Medium
Checkout and payment	Large
Wishlist	Medium
Order tracking	Medium

Points: Small 1, Medium 2, Large 3. Sprint capacity is 5 points. Write a one-line Sprint Goal and one QA tas

SAMPLE ANSWER

Pick: Search (1) + Login (1) + Add to cart (2) = 4 points. This fits with a buffer. Checkout (3) plus Add to cart (2) = 5 would also fit.

Goal: "A signed-in shopper can find a product and add it to the cart."

QA task: write tests for the cart from the acceptance criteria.

Many answers work. Check that the goal is clear, fits capacity and ends in a testable, working increment.

3 min

Quick quiz: show A, B, C or D

QUESTION 1

Which statement is correct?

A Agile and Scrum are the same

B Scrum is a framework that follows Agile

C Agile is a Scrum role

D Scrum is older than Agile

QUESTION 2

Who decides the order of items in the Product Backlog?

A Scrum Master

B Product Owner

C QA tester

D Project sponsor only

QUESTION 3

When does QA test in an Agile sprint?

A Only after the sprint ends

B Only on the last day

C Throughout the sprint

D Only before release

B — a framework

Agile is the mindset; Scrum is one way to practise it.

B — Product Owner

They own and order the backlog by value.

C — Throughout

Test each story as soon as it is ready.

1 min

Five things to remember

01 Agile is a mindset: small pieces, early feedback, adapt to change.

02 Scrum is one framework that follows Agile. They are not the same thing.

03 Three roles, four events, two backlogs, one sprint at a time.

04 Each sprint ends with a working, tested increment.

05 QA works inside the sprint, from refinement to regression.

Module 02 complete

- ✓ Agile and Waterfall
- ✓ Scrum roles and events
- ✓ Backlogs and sprints
- ✓ QA in Agile

Up next: User stories

MODULE 03 OF 03 · 30 MINUTES

User stories and acceptance criteria …

What users need, and how QA turns it into tests

OBJECTIVE 1

Write a user story in the standard format

OBJECTIVE 2

Write clear, testable acceptance criteria

OBJECTIVE 3

Turn a story into positive and negative test scenarios

4 min

From requirements to user stories

Software requirements

Statements of what the software must do (functional) and how well it must do it (non-functional, such as speed and security).

User story

A short, plain-language description of a feature from the user's point of view. The start of a conversation, not a full specification.

```
As a [type of user],
I want [functionality],
So that [benefit].
```

WHY TEAMS USE USER STORIES

They keep the focus on the user's need.

Everyone, technical or not, can read them.

They are small enough to finish in one sprint.

4 min

Three parts: who, what, why

AS A …

The user

Who needs this? A real role, such as shopper or admin.

I WANT …

The function

What should the software let them do?

SO THAT …

The benefit

Why does it matter? The value to the user.

SHOPEASY EXAMPLE: LOGIN

As a registered shopper, **I want** to log in with my email and password, **so that** I can see my orders and check out quickly.

Weak story: “Make a login page.” No user, no purpose, nothing to test against.

Good story: clear user, clear need, clear value, and small enough for one sprint.

3 min

Acceptance criteria say when the story is done

User story

The need: **who, what, why**. Short and in plain words.

Acceptance criteria

The conditions that must be true for the story to be accepted. Specific and testable.

LOGIN: ACCEPTANCE CRITERIA

1. A shopper with a valid email and password reaches their account page.
2. A wrong password shows “Incorrect email or password” and stays on the login page.
3. An empty email or password field shows a message that the field is required.
4. After 5 failed attempts the account is locked for 15 minutes.

Good criteria are clear, specific, testable (pass or fail) and written in simple words. Not: “Login should be fast and easy.”

3 min

How QA turns a story into tests

STEP 1
User story
Understand who needs what and why.

→

STEP 2
Acceptance criteria
Read each pass or fail condition.

→

STEP 3
Test scenarios
List what to test, in one line each.

→

STEP 4
Test cases
Steps, data and expected result.

Acceptance criterion	Test scenario	Test case (expected result)
Valid login	Log in with a valid email and password	User sees the account page
Wrong password	Log in with a wrong password	Error message; stays on login
Empty fields	Submit with email left blank	“Email is required” is shown

Scenario = what to test (short). **Test case** = how to test it: steps, test data and expected result.

3 min

Positive and negative test scenarios

Positive

Valid input, expected use. Checks that the feature works as intended.

Log in with a valid email and password.
Add an in-stock item to the cart.

Negative

Invalid input or misuse. Checks that the software handles errors safely and clearly.

Log in with a wrong password.
Add an out-of-stock item to the cart.

IDEAS FOR FINDING NEGATIVE SCENARIOS

Empty or missing data · too long or too short · wrong format · special characters · wrong order of steps · network lost · using the feature while logged out

Both types are needed. Positive tests show it works. Negative tests show it does not break.

4 min

Worked example: Add to Cart

As a shopper, **I want** to add a product to my cart, **so that** I can buy several items in one order.

ACCEPTANCE CRITERIA

1. Add to Cart is shown on in-stock products.
2. Clicking it adds one item and the cart count goes up by one.
3. Adding the same item again raises its quantity.
4. Out-of-stock products cannot be added.
5. The cart total updates immediately.

POSITIVE SCENARIOS

Add one in-stock item; count becomes 1.

Add the same item twice; quantity is 2.

Add two different items; total is the sum.

NEGATIVE SCENARIOS

Try to add an out-of-stock item.

Add more than the available stock.

Double-click Add to Cart quickly.

Add an item with the network off.

4 min

Story to scenarios: Search

As a shopper, **I want** to search for products by name, **so that** I can find what I need qui

In pairs, spend four minutes on three tasks:

1. Write three acceptance criteria.
2. Write two positive and three negative test scenarios.
3. Mark each scenario as positive or negative.

Hint: think of empty search, no results, spelling mistakes, very long text, special characters.

SAMPLE ANSWER

Criteria: (1) Results show products whose name matches the text. (2) A search with no match shows “No products found”. (3) Results appear within 3 seconds.

Positive: search “phone” and see phones; search a full product name and see that product.

Negative: empty search box; a word with no match; special characters such as @#$; a very long text; a spelling mistake.

Any clear, testable answer is valid.

3 min

Quick quiz: show A, B, C or D

QUESTION 1

What does “So that…” in a user story give?

A The user’s role

B The test steps

C The benefit to the user

D The code design

QUESTION 2

Which is a testable acceptance criterion?

A Checkout should be easy

B The page must be fast

C Looks nice on phones

D Order confirmation shows after payment

QUESTION 3

“Log in with a wrong password” is which kind of scenario?

A Positive

B Negative

C A user story

D A requirement

C — The benefit

It explains why the user wants the feature.

D — Specific and checkable

The others are vague and cannot be marked pass or fail.

B — Negative

Invalid input; the system must show an error.

1 min

The whole programme in one page

01 · SDLC

Six phases from planning to maintenance. QA adds value in every phase.

02 · AGILE

Agile is the mindset. Scrum is one framework. QA works inside each sprint.

03 · USER STORIES

Story = the need. Criteria = the checklist. QA turns both into scenarios.

MODULE 3 KEY POINTS

Format: As a [user], I want [function], so that [benefit].

Good criteria are specific and testable (pass or fail).

Write positive and negative scenarios; ask about anything unclear.

All three modules complete